Worksheet 23

November 30, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Review: an angle-sum exact value

Find $\cos 165°$ exactly, without a calculator.

(a) Write the angle as a sum of two familiar angles.

(b) Apply the cosine sum formula and simplify the exact value.

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2. Review: double angles

Suppose $\cos\theta = \frac{3}{5}$ and $\frac{3\pi}{2} < \theta < 2\pi$.

(a) Find the exact value of $\sin\theta$, using the quadrant to choose its sign.

(b) Find $\sin(2\theta)$, $\cos(2\theta)$, and $\tan(2\theta)$ exactly.